Mohammad Pivezhandi |

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Summary

Assistant Professor (Teaching) of Computer Science at Wayne State University and Researcher in the Vehicle Health group at the General Motors R&D Center. Ph.D. in Computer Science (Wayne State University, December 2025, GPA 4.0/4.0) specializing in AI-guided energy- and thermal-aware scheduling, reinforcement learning, and reconfigurable computing. First-author and co-author of 10+ papers including Ph.D. dissertation and ArXiv preprints in top-tier venues (e.g., IJCAI-ECAI, RTSS, RTCSA, ECRTS). Proficient in Python, TensorFlow, PyTorch, VHDL, and FPGA design, with extensive research and teaching experience in academia and industry.

U.S. permanent resident (green card). No visa sponsorship required.

Skills

Programming: MATLAB, Python, C, C++, Java, MIPS Assembly, Shell, R

ML/AI Frameworks: TensorFlow, PyTorch, CUDA, OpenCV, OpenMP, MPI, PGAS

Hardware Design: VHDL, Verilog HDL, SystemC

Simulation Tools: Proteus, ModelSim, Active HDL, Spice

CAD/EDA: Virtuoso, Proteus, Altium Designer, OrCAD, Microsoft Visio

Productivity: LaTeX, Microsoft Office, Eclipse, NetBeans, Vivado HLS, Vivado HLx, AMS, Quartus, Unix, Git, Slack

Languages: English (Fluent), Persian (Native)

Experience

Wayne State University — **Detroit, MI**
Assistant Professor (Teaching), Computer Science — *Aug 2026–Present*

- Full-time, non-tenure-track faculty appointment in the Department of Computer Science, teaching within the online Artificial Intelligence degree program.
- Instructor for Software Engineering (CSC 4110) and AI Literacy (CSC 1001), Fall 2026.
- Contributing to curriculum development, assessment, student recruitment, and departmental service.

General Motors R&D Center — **Warren, MI**
Researcher, Vehicle Health Group (Contractor) — *Apr 2026–Present*

- Developing AI/ML-based solutions for vehicle health management and predictive diagnostics in energy and propulsion systems at the GM Research & Development Center.
- Applying physics-based modeling and machine learning techniques to analyze large-scale vehicle telemetry data for fault detection and performance optimization.
- Continuing on a reduced, part-time schedule since August 2026, concurrent with the Wayne State University faculty appointment.

Wayne State University — **Detroit, MI**
Graduate Research Assistant, Computer Science — *2022–2025*

- Engineered a Hierarchical Multi-Agent Reinforcement Learning (HMARL) framework for energy- and performance-aware scheduling of OpenMP DAG workloads on heterogeneous multicore platforms (NVIDIA Jetson TX2, Intel Core i7), achieving 49.06% makespan reduction and 40.95% energy reduction compared to state-of-the-art multi-agent baselines; published as RTSS 2024 WIP.
- Designed a statistical learning framework using Random Forest, backward stepwise regression, and Pearson correlation for feature-aware task-to-core allocation, reducing energy by 10% and core temperature by 5°C; published at RTCSA 2025.

Iowa State University **Ames, IA**
Graduate Research Assistant, Computer Science *2021–2022*

- Architected energy-aware scheduling algorithms for DAG-based inputs on Intel Xeon processors in data-processing clusters, improving job completion time by 15% via a dynamic resource allocation simulator; published at PDCAT 2021.
- Developed a reinforcement learning-based scheduler using Graph Neural Networks (GNNs) for high-volume ETL processing in data warehouse systems, achieving up to 15% improvement in job completion time; published at PDCAT 2021.
- Designed an FPGA-based architecture for histogram generation to support event-based camera optical flow calculation, reducing time encoding from 32 to 7 bits with $\sim$3% loss in histogram accuracy; published at ASAP 2020.

Moffett Systems, Inc. **Los Altos, CA**
Research Intern, Full-Stack Hardware Accelerator Design *2020–2021*

- Optimized FPGA prototyping, RTL design, compiler development, firmware, and runtime for hardware accelerators, reducing verification latency by 15% through debugging and bring-up enhancements.

Iowa State University **Ames, IA**
Graduate Teaching Assistant, Electrical and Computer Engineering *2019–2022*

- Mentored 180+ undergraduate students in Advanced Object-Oriented Programming (CprE 327, Fall/Spring 2020), teaching algorithms, data structures, and advanced C programming through tutoring sessions and exam proctoring.
- Guided 140+ undergraduate students in Introduction to C Programming (CprE 185, Fall 2021), covering pointers, memory management, and file I/O, enhancing comprehension via weekly tutoring sessions.
- Served as course instructor for Computer Organization (CprE 381, Spring 2022, 200+ undergraduate students), teaching 32-bit pipelined MIPS processor implementation in VHDL and Assembly, developing course materials and managing all instructional responsibilities.

Mashhad Researchers Center **Mashhad, Iran**
FPGA/ASIC Design Researcher *2015–2018*

- Developed high-throughput matrix inversions for MIMO wireless applications using the Gauss-Jordan algorithm on FPGA, improving processing speed by 20%.
- Implemented fault-tolerant systems on FPGA using HDL and SystemC, achieving 99% reliability through MATLAB-based simulations.
- Optimized AES cryptography algorithm on Spartan-6 FPGA with RTL design, reducing area by 12%; simulated using ActiveHDL and ISE Design Suite.

Shahid Beheshti University **Tehran, Iran**
Graduate Research and Teaching Assistant, Digital Electronics *2013–2015*

- Designed a low-power, high-frequency ASIC layout for an 8-bit multiplier using pseudo-nMOS and pseudo-pMOS gates, reducing power consumption by 18%.
- Developed statistical-based models for CNT-FET applications using support vector regression and random forest, achieving 9% error for off-state current; published in Springer J. Comput. Electron. 2017.
- Designed an ASIP for CORDIC-based DFT and DCT algorithms, reducing gate count by 84% for DFT and 86% for DCT; published at ICCKE 2016.
- Built a statistical-based neural network for human activity recognition, achieving over 99% accuracy; published in Int. J. Comput. Appl. 2015.
- Taught Machine Learning course for the first time in the department to 50+ undergraduate and graduate students, developing all lecture content, assignments, and course materials using MATLAB and Python from scratch.

Education

Computer Science — **Wayne State University**
Doctor of Philosophy (Ph.D.), Detroit, MI — *2022–2025*
CGPA: 4.0/4.0

Dissertation: *Data-Efficient AI-Guided Energy- and Thermal-Aware Scheduling on Heterogeneous Multicore Systems* (December 2025)

Dissertation Link: https://github.com/pivezhan/PhD-Thesis-Public

Computer Science/Engineering — **Iowa State University**
Master of Science (M.Sc.), Ames, IA — *2019–2022*
CGPA: 3.76/4.0

Master's Thesis: *Toward Real-Time Energy-Aware Automation of Resource Scheduling Using Reinforcement Learning*

Digital Electronics — **Shahid Beheshti University**
Master of Science (M.Sc.), Tehran, Iran — *2013–2015*
CGPA: 4.0/4.0

Master's Thesis: *Statistical Analysis, Design, and Implementation of Artificial Neural Networks for Human Activity Recognition Task (Nominated as Best Research in ECE Department)*

Electronic Engineering — **Ferdowsi University of Mashhad**
Bachelor of Science (B.Sc.), Mashhad, Iran — *2008–2013*
Last Two Semesters: 16.68/20

Bachelor's Thesis: *Design and Implementation of a Mobile Communication System Using Microcontrollers (Nominated as Best Bachelor's Final Project)*

Publications

Ph.D. Dissertation

1. **Mohammad Pivezhandi**, "Data-Efficient AI-Guided Energy- and Thermal-Aware Scheduling on Heterogeneous Multicore Systems," *Ph.D. Dissertation, Wayne State University*, December 2025.

Preprints (ArXiv)

1. **Mohammad Pivezhandi**, A. Saifullah, A. Jannesari, "HiDVFS: A Hierarchical Multi-Agent DVFS Scheduler for OpenMP DAG Workloads," *arXiv preprint arXiv:2601.06425*, 2026.
2. **Mohammad Pivezhandi**, M. Banisharif, A. Saifullah, A. Jannesari, "ZeroDVFS: Zero-Shot LLM-Guided Core and Frequency Allocation for Embedded Platforms," *arXiv preprint arXiv:2601.08166*, 2026.
3. **Mohammad Pivezhandi**, A. Saifullah, P. Modekurthy, "Feature-Aware Task-to-Core Allocation in Embedded Multi-core Platforms via Statistical Learning," *arXiv preprint arXiv:2502.15716*, 2026.
4. **Mohammad Pivezhandi**, A. Saifullah, "FlowRL: Flow-Augmented Few-Shot Reinforcement Learning for Semi-Structured Sensor Data," *arXiv preprint arXiv:2409.14178*, 2024.

Journal Publications

1. **Mohammad Pivezhandi**, K. Abedi, A. Hassanzadeh, "Accuracy Improvement with Reliable Statistical-Based Models for CNT-FET Applications," *Springer Journal of Computational Electronics*, 2017.
2. **Mohammad Pivezhandi**, B. M.-N. Maybodi, "Statistical Based Neural Network in Human Activity Recognition," *International Journal of Computer Applications*, vol. 124, no. 12, 2015.

Conference Publications

1. **Mohammad Pivezhandi**, M. Banisharif, S. Bakhshan, A. Saifullah, A. Jannesari, "GraphPerf-RT: Graph-Driven Performance Modeling with Calibrated Uncertainty for OpenMP Scheduling on Heterogeneous Embedded SoCs," *International Joint Conference on Artificial Intelligence (IJCAI-ECAI), Special Track on AI4Tech: AI Enabling Critical Technologies*, 2026.
2. **Mohammad Pivezhandi**, M. Banisharif, A. Saifullah, "ZeroDVFS: Zero-Shot LLM-Guided Autonomous Agent for Energy-Aware Resource Allocation in Embedded Systems," *ICLR Workshop on Agentic AI in*

the Wild: From Hallucinations to Reliable Autonomy, 2026.

3. **Mohammad Pivezhandi**, A. Saifullah, P. Modekurthy, "Feature-Aware Task-to-Core Allocation in Embedded Multi-core Platforms via Statistical Learning," *IEEE Real-Time Computing and Applications Symposium* (*RTCSA*), 2025.
4. **Mohammad Pivezhandi**, A. Jain, A. Saifullah, A. Jannesari, "Energy and Thermal-Aware Scheduling based on HMARL for OpenMP DAG Workloads," *Work-In-Progress, IEEE Real-Time Systems Symposium* (*RTSS*), 2024.
5. A. Ahmed, **Mohammad Pivezhandi**, Z. Guo, J. Li, V. P. Modekurthy, A. Saifullah, "Precise Scheduling of DAG Tasks with Dynamic Power Management," *Euromicro Conference on Real-Time Systems* (*ECRTS*), 2023.
6. Q. Bashir, **Mohammad Pivezhandi**, A. Saifullah, "Energy- and Temperature-aware Scheduling: From Theory to Practical Implementation on Intel Processor," *IEEE International Conference on High Performance Computing and Communications* (*HPCC*), 2022.
7. V. Bengre, M. R. HoseinyFarahabady, **Mohammad Pivezhandi**, A. Y. Zomaya, A. Jannesari, "A Learning-Based Scheduler for High Volume Processing in Data Warehouse using Graph Neural Networks," *International Conference on Parallel and Distributed Computing* (*PDCAT*), 2021.
8. **Mohammad Pivezhandi**, P. H. Jones, J. Zambreno, "ParaHist: FPGA Implementation of Parallel Event-Based Histogram for Optical Flow Calculation," *IEEE International Conference on Application-specific Systems, Architectures and Processors* (*ASAP*), 2020.
9. **Mohammad Pivezhandi**, M. Eshghi, "ASIP Design for Two Dimensional Cordic based DFT and DCT Algorithms," *International Conference on Computer and Knowledge Engineering* (*ICCKE*), 2016.

Honors

Aug 2026: Presented *GraphPerf-RT* at the International Joint Conference on Artificial Intelligence (IJCAI-ECAI 2026), Special Track on AI4Tech: AI Enabling Critical Technologies, Bremen, Germany.

May 2025: Awarded a travel grant to attend the IEEE Real-Time Computing and Applications Symposium (RTCSA) in Singapore, recognizing contributions to energy-aware scheduling research.

2022–2026: Invited as peer reviewer for prestigious journals and conferences, including ACM Transactions on Embedded Computing Systems (TECS), Conference on Neural Information Processing Systems (NeurIPS 2026), Association for the Advancement of Artificial Intelligence (AAAI 2026), and Artificial Intelligence and Statistics (AISTATS). Served as TPC member in International Symposium on Parallel and Distributed Processing with Applications (ISPA) and International Wireless Communications and Mobile Computing Conference (IWCMC) 2026 Machine Learning Symposium.

2022–2025: Secured National Science Foundation (NSF) funding for research on AI-guided energy- and thermal-aware scheduling for heterogeneous multicore systems.

2016–2026: Published as first or second author in prestigious conferences, including IJCAI-ECAI, RTSS, RTCSA, ECRTS, HPCC, PDCAT, ASAP, and ICCKE.

Sep 2018: Received Ph.D. admission offers from top universities in the USA, Europe, Australia, Canada, and Iran.

Sep 2015: M.Sc. Thesis nominated as Best Research in the Shahid Beheshti University ECE Department.

Sep 2013: Senior design project nominated as Best Bachelor's Final Project at Ferdowsi University of Mashhad.

Aug 2008: Ranked in the 95th percentile in International Physics and Mathematics exam.

Aug 2008: Ranked in the 99th percentile in National Azad University entrance exam in Physics and Mathematics.

Courses

Graduate and Undergraduate Courses

- Deep Learning (Iowa State University, A, Dec 2020)
- Computational Perception (Iowa State University, A, May 2020)

- High Performance Computing (Iowa State University, A, Dec 2019)
- Application of Parallel Computers (University of California, Berkeley, A, May 2019)
- Probabilistic Methods and Algorithms (Iowa State University, A, May 2019)
- Linear Integrated Circuit Designs (Shahid Beheshti University, 17.25/20, Mar 2015)
- Semiconductor Devices I (Shahid Beheshti University, 18.2/20, Mar 2015)
- Application Specific Instruction Set Processor Designs (ASIP) (Shahid Beheshti University, 16.5/20, Aug 2014)
- Digital Image Processing (Shahid Beheshti University, 16.62/20, Aug 2014)
- VLSI Circuits Design (Shahid Beheshti University, 18.6/20, Mar 2014)
- Automatic Digital Circuit Designs with VHDL (Shahid Beheshti University, 18/20, Mar 2014)

Certificates

- Introduction to Software Engineering (IBM, 90%)
- Algorithmic Toolbox (University of California San Diego, 67.51%)
- Introduction to Cloud Computing (IBM, 89.50%)
- Java Programming: Solving Problems with Software (Duke University, 94.15%)
- Coding Interview Preparation (Meta, 95%)
- Generative AI: Elevate your Software Development Career (IBM, 87%)
- Software Developer Career Guide & Interview Preparation (IBM, 73.40%)
- Python for Data Science, AI & Development (IBM, 94.75%)
- LangChain for LLM Application Development (DeepLearning.AI, 100%)
- Generative AI with Large Language Models (DeepLearning.AI, 92.50%)
- Data Structures & Algorithms Specialization (University of California San Diego)
- Deep Learning Specialization (Stanford University / DeepLearning.AI)
- Data Science Specialization (Johns Hopkins University)
- Machine Learning (Stanford University, 99.3%)
- The Data Scientist's Toolbox (Johns Hopkins University, 99.3%)
- R Programming (Johns Hopkins University, 100%)
- Getting and Cleaning Data (Johns Hopkins University, 98%)
- Statistical Inference (Johns Hopkins University, 100%)
- Reproducible Research (Johns Hopkins University, 97.1%)
- Regression Models (Johns Hopkins University, 91.7%)
- Exploratory Data Analysis (Johns Hopkins University, 96.7%)
- Practical Machine Learning (Johns Hopkins University, 100%)
- Developing Data Products (Johns Hopkins University, 96.9%)

Hobbies

- Aerial Shooting
- Guitar Playing
- Chess
- Cooking
- Running
- Swimming

References

- Abusayeed Saifullah, Associate Professor, The University of Texas at Dallas, Abusayeed.Saifullah@UTDallas.edu
- Prashant Modekurthy, Assistant Professor, University of Nevada Las Vegas, prashant.modekurthy@unlv.edu
- Ali Jannesari, Associate Professor, Iowa State University, jannesar@iastate.edu